

Probability Notes

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Chapter 1 Probability Space

Definition 1.1 (Probability Space)

Probability space is a tuple $(\Omega,\ \mathcal{F},\ \mathbf{P})$ *where*

- *Sample space* Ω *is a set of all possible outcomes.*
- σ*-algebra* $\mathcal{F}$ *is a collection of subsets of* Ω*.*
- *Probability measure* $\mathbf{P}$ *is a function that assigns a probability to each event in* $\mathcal{F}$*.*

♣

Definition 1.2 (Sample Space)

Sample space Ω *is a set of all possible outcomes, which should satisfy the following properties:*

- *Mutually Exclusive: The elements in* Ω *should be unique.*
- *Collectively Exhaustive: The elements in* Ω *should cover all possible outcomes.*

♣

Once the experiment is conducted, there is **exactly one** element in the sample space Ω occurs.

Example 1.1 (Example of Sample Space) Here are some examples of sample space:

- Discrete: $\Omega = \{1, 2, 3, 4, 5, 6\}$ - Rolling a fair die.
- Continuous: $\Omega = [0, 1]$ - Randomly selecting a real number between 0 and 1.

Before we define the σ-algebra, we need to introduce the concept of **countable**. We say a set X is countable if there is a bijection between X and $\mathbb{N}$. Some countable sets are: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{N}^2$, $\mathbb{Z}^2$, $\mathbb{Q}^2$, $\cup_{n=1} X_n$. Some uncountable sets are: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{R}^2$, $\mathbb{C}^2$.

Definition 1.3 (σ-algebra)

We say a set $\mathcal{F}$ *collection of subsets of* Ω *is a* σ*-algebra if it satisfies the following properties:*

♣

Remark Not all the things in real worlf can be measured by probability, especially when things are continuous (uncountable).

Chapter 2 Conditional Probability

2.1 Conditional Probability

Definition 2.1 (Conditional Probability)

Conditional probability is the probability of an event A given that another event B has already occurred. It is denoted by $P(A|B)$ and is defined as:

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} \tag{2.1.1}$$

Note that when $\mathbf{P}(B) = 0$, the conditional probability is undefined.

♣

We can easily check the conditional probability satisfies the properties of probability measure, which means it is a legitimate probability on a new universe.

- Non-negativity: $\mathbf{P}(A|B) \geq 0$.
- Normalization: $\mathbf{P}(\Omega|B) = 1$.
- Additivity: when A and B are disjoint, $\mathbf{P}(A \cup B|C) = \frac{\mathbf{P}((A\cup B)\cap C)}{\mathbf{P}(C)} = \frac{\mathbf{P}(A\cap C)+\mathbf{P}(B\cap C)}{\mathbf{P}(C)} = \mathbf{P}(A|C) + \mathbf{P}(B|C)$. (The second equality is due to $A \cap C$ and $B \cap C$ are disjoint).

Example 2.1 (Discrete and Continuous) For discrete case, the conditional probability can be calulated by:

$$\mathbf{P}(A|B) = \frac{\#\ of\ elements\ in\ A}{\#\ of\ elements\ in\ B} \tag{2.1.2}$$

For continuous case, the conditional probability can be calculated by:

$$\mathbf{P}(A|B) = \frac{area\ of\ A}{area\ of\ B} \tag{2.1.3}$$

When solving a problem, the following equations may help:

$$\mathbf{P}(A \cap D) = \mathbf{P}(A) \cdot \mathbf{P}(D|A) = \mathbf{P}(D) \cdot \mathbf{P}(A|D) \tag{2.1.4}$$

$$\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B|A) \cdot \mathbf{P}(C|A \cap B) \tag{2.1.5}$$

For the second equation: 1.event A occurs; 2.event B occurs given A; 3.event C occurs given A and B. Or

$$\begin{aligned}\mathbf{P}(A \cap B \cap C) &= \mathbf{P}(A \cap B) \cdot \mathbf{P}(C|A \cap B) \\ &= \mathbf{P}(A) \cdot \mathbf{P}(B|A) \cdot \mathbf{P}(C|A \cap B)\end{aligned}$$

Remark These equations are not fixed, depending on $\mathbf{P}(A)$, $\mathbf{P}(\mathbf{A} \cap \mathbf{B})$ and $\mathbf{P}(A|B)$ which are easier to calculate.

2.2 Total Probability Theorem

We can calculate the total probability with *Devide and Conquer* strategy:

1. Devide event B by $A_1, A_2, \cdots A_n$, note that $\bigcup_{i=1}^{n} A_i = \Omega$ and $A_i \cap A_j = \emptyset$.
2. Calculate $\mathbf{P}(B|A_i)$ which are easier to calculate.
3. Sum all the probabilities.

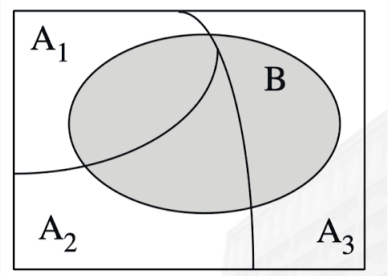


Figure 2.1: Total Probability Theorem

Theorem 2.1 (Total Probability Theorem)

Let $A_1, A_2, \cdots, A_n$ be a partition of the sample space Ω. Then for any event B,

$$\mathbf{P}(B) = \sum_{i=1}^{n} \mathbf{P}(A_i) \cdot \mathbf{P}(B|A_i) \tag{2.2.1}$$

♡

Example 2.2 (Die Rolling) You roll a fair four-sided die. If the result is 1 or 2, you roll once more but otherwise, you stop. What is the probability that the sum total of rolls is at least 4?

Solution

1. *Let* $A_i = \{the\ first\ roll\ is\ i\}, B = \{the\ sum\ total\ of\ rolls\ is\ at\ least\ 4\}$.
2. $\mathbf{P}(A_1) = \mathbf{P}(A_2) = \mathbf{P}(A_3) = \mathbf{P}(A_4) = \frac{1}{4}$.
3. $\mathbf{P}(B|A_1) = \frac{1}{2}, \mathbf{P}(B|A_2) = \frac{3}{4}, \mathbf{P}(B|A_3) = 0, \mathbf{P}(B|A_4) = 1$.
4. $\mathbf{P}(B) = \frac{9}{16}$.

2.3 Bayes' Theorem

If we know:

- "Prior" probabilities $\mathbf{P}(A_i)$.
- "Likelihood" probabilities $\mathbf{P}(B|A_i)$.

We wish we can calculate the "Posterior" probabilities $\mathbf{P}(A_i|B)$.

Theorem 2.2 (Bayes' Theorem)

Let $A_1, A_2, \cdots, A_n$ be a partition of the sample space Ω. Then for any event B

$$\begin{aligned} \mathbf{P}(A_i|B) &= \frac{\mathbf{P}(A_i \cap B)}{\mathbf{P}(B)} \\ &= \frac{\mathbf{P}(A_i) \cdot \mathbf{P}(B|A_i)}{\sum_{j=1}^{n} \mathbf{P}(A_j) \cdot \mathbf{P}(B|A_j)} \\ &= \frac{\mathbf{P}(A_i) \cdot \mathbf{P}(B|A_i)}{\sum_{j=1}^{n} \mathbf{P}(A_j) \cdot \mathbf{P}(B|A_j)} \end{aligned} \tag{2.3.1}$$

♡

A: cause, B: result. Bayes' Theorem is used to calculate the cause given the result. *(inference based on probability)*

Chapter 3 Independence

3.1 Independence

Definition 3.1 (Independence)

Event A and B are independent if:

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B) \tag{3.1.1}$$

or equivalently, when $\mathbf{P}(B) \neq 0$:

$$\mathbf{P}(A|B) = \mathbf{P}(A) \tag{3.1.2}$$

♣

Occurrence of B does not provides no information about the occurrence of A.

Remark Disjoint events are not independent.

$$\begin{aligned} \mathbf{P}(A \cap B) &= 0 \\ \mathbf{P}(A \cap B) &= \mathbf{P}(A) \cdot \mathbf{P}(B) \end{aligned} \tag{3.1.3}$$

Only if $\mathbf{P}(A) = 0$ or $\mathbf{P}(B) = 0$, the two equations can be both satisfied. E.g. when a student is in one class, it is unlikely that he is in another class.

3.2 Conditional Independence

Definition 3.2 (Conditional Independence)

Event A and B are conditionally independent given C if:

$$\mathbf{P}(A \cap B|C) = \mathbf{P}(A|C) \cdot \mathbf{P}(B|C) \tag{3.2.1}$$

♣

Remark Conditional independence does not imply independence. Information of the mainland of China does not imply the information of Tsinghua University.

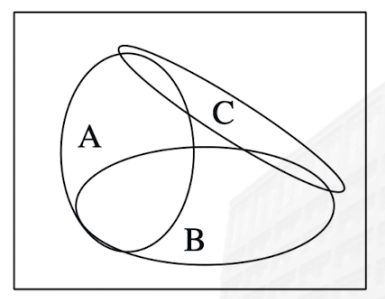


Figure 3.1: Conditional Independence

As shown in Figure 3.1, assume A and B are independent. Obviously, A and B are not independent given C (they are disjoint in C).

3.3 Independence of Multiple Events

Definition 3.3 (Independence of multiple events)

Events $A_1, A_2, \cdots, A_n$ *are independent if*

$$\mathbf{P}(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = \mathbf{P}(A_{i_1}) \cdot \mathbf{P}(A_{i_2}) \cdots \mathbf{P}(A_{i_k}) \tag{3.3.1}$$

for any subset $\{i_1, i_2, \cdots, i_k\} \subseteq \{A_1, A_2, \cdots, A_n\}$.

♣

Remark Pair independence does not imply independence of multiple events. We should check if $\mathbf{P}(\bigcap_{i=1}^{n} A_i) = \prod_{i=1}^{n} \mathbf{P}(A_i)$.

Chapter 4 Discrete Random Variables

4.1 Random Variables

Random variables (RV) is not a variable, but a diterministic function that maps the sample space to the real number, whose range is either finite or countably infinite. Notation: $X(\omega) = x$.

Definition 4.1 (Random Variable)

Given a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, a random variable X is a function $X : \Omega \to \mathbb{R}$ such that for any $x \in \mathbb{R}$, the set $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$.

♣

The intuition of introducing RV is to map the sample space to $\mathbb{R}$ so that we can apply some mathmatical tools to analyze the probability distribution of the sample space. But the probability meature on $\mathbb{R}$ is defined on Borel field $\mathcal{B}(\mathbb{R})$, **so if we want to calculate the probability of RV**, we should consider the a set $A \in \mathcal{B}(\mathbb{R})$. To maintain the consistency between RV and events, the pre-image of A under X should be in $\mathcal{F}$, i.e. for any $x \in \mathbb{R}$, the set $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$.
Remark The requirement $\forall x \in \mathbb{R},\ \{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ is naturally met when the events in Ω are discrete, because we can choose $\mathcal{F} = 2^{\Omega}$.

Example 4.1 (Examples of RV)

A sequence of 5 tosses of a coin:

- The number of heads in the sequence [Discrete RV]
- The 5-long sequence is not a RV.

Two rolls of a die:

- The sum of two rolls [Discrete RV]
- The second roll to the power of the first roll [Discrete RV]

DeepSeek's inference:

- The latency in generating a response [Continuous RV]
- The number of requests in a second [Discrete RV]

4.2 Probability Mass Function

Definition 4.2 (Probability Mass Function)

*Probability mass function (PMF) is the "probability law" or "probability distribution" of a **discrete RV** X.*

$$p_X(x) = \mathbf{P}(X = x) = \tag{4.2.1}$$

$$= \mathbf{P}(\{\omega \in \Omega : X(\omega) = x\}) \tag{4.2.2}$$

♣

Several properties of PMF can be derived from the axioms of probability, e.g. $p_X(x) \geq 0$, $\sum_x p_X(x) = 1$.

Example 4.2 (Binomial PMF)

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \tag{4.2.3}$$

One example: the number of heads in n independent tosses of a coin.

Example 4.3 (Geometric PMF)

$$p_X(k) = (1-p)^{k-1} p \tag{4.2.4}$$

One example: the number of tosses until the first head appears.

Example 4.4 (Poisson PMF)

$$p_X(k) = \mathrm{e}^{-\lambda} \frac{\lambda^k}{k!} \tag{4.2.5}$$

It's a good approximation of the binomial distribution when n is large and p is small.

Proof Let $p = \frac{\lambda}{n}$, then

$$\begin{aligned}
\binom{n}{k} p^k (1-p)^{n-k} &= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \\
&= \frac{n(n-1)\cdots(n-k+1)}{k!} p^k (1-p)^{n-k} \\
&\approx \frac{n^k}{k!} p^k (1-p)^{n-k} \\
&= \frac{n^k}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^n \\
&\approx \frac{n^k}{k!} \left(\frac{\lambda}{n}\right)^k \mathrm{e}^{-\lambda} \\
&= \mathrm{e}^{-\lambda} \frac{\lambda^k}{k!}
\end{aligned} \tag{4.2.6}$$

Note that all these PMFs are normalized.

4.3 Functions of Random Variables

Definition 4.3 (Function of Random Variables)

Given a RV X, a function $Y = g(X)$ is also a RV. If X is discrete, then Y is also discrete, with PMF p_Y given by

$$p_Y(y) = \sum_{x:g(x)=y} p_X(x) \tag{4.3.1}$$

♣

Remark The function g is defined on $\mathbb{R}$, i.e. $g : \mathbb{R} \to \mathbb{R}$. The meaning of $Y = g(X)$ in fact is $Y(\omega) = g(X(\omega))$.

4.4 Expectation and Variance

Definition 4.4 (Expectation of Random Variables)

The expected value (also called the expectation or the mean) of a random variable X, with PMF p_X, is defined as

$$\mathbf{E}[X] = \sum_x x p_X(x) \tag{4.4.1}$$

♣

While the PMF gives us the full information about the distribution of X, sometimes we still need an overall measure of X.

Property 4.1 (Properties of Expectation)

- *Let X be a RV and $Y = g(X)$. Sometimes $\mathbf{E}[Y] = \sum_y y p_Y(y)$ is hard to calculate. Instead, we can use $\mathbf{E}[Y] = \sum_x g(x) p_X(x)$ which is easier.*

  Proof

$$\begin{aligned}
\mathbf{E}[Y] &= \sum_y y p_Y(y) && (4.4.2)\\
&= \sum_y y \sum_{x:g(x)=y} p_X(x) && (4.4.3)\\
&= \sum_x g(x) p_X(x) && (4.4.4)
\end{aligned}$$

- *In general, $\mathbf{E}[g(X)] \neq g(\mathbf{E}[X])$.*
- *If $\alpha_1, \alpha_2, \beta$ are constants, X, Y are RVs, then*

$$\mathbf{E}[\alpha_1 X + \alpha_2 Y + \beta] = \alpha_1 \mathbf{E}[X] + \alpha_2 \mathbf{E}[Y] + \beta \tag{4.4.5}$$

Definition 4.5 (Variance of Random Variables)

The variance of a RV X is defined as

$$\operatorname{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] \tag{4.4.6}$$

Remember $\mathbf{E}[g(x)] = \sum_x g(x)p_X(x)$, then $\operatorname{var}(X) = \sum_x (x - \mathbf{E}[X])^2 p_X(x)$. We can also define the standard deviation as $\sigma(X) = \sqrt{\operatorname{var}(X)}$.

Property 4.2 (Properties of Variance)

- *Define second moment as* $\mathbf{E}[X^2] = \sum_x x^2 p_X(x)$*, then* $\operatorname{var}(X)$ *can be written in terms of moments expression* $\operatorname{var}(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2$.
- $\operatorname{var}(X) \geq 0$.
- $\operatorname{var}(\alpha X + \beta) = \alpha^2 \operatorname{var}(X)$.

Example 4.5 (Expectation and Variance of Binomial Distribution)

$$\begin{aligned}
\mathbf{E}[X] &= \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=1}^{n} np \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
&= np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
&= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^{k} (1-p)^{n-1-k} \\
&= np \cdot \mathrm{PMF} = np
\end{aligned} \tag{4.4.7}$$

$$\begin{aligned}
\mathbf{E}[X^2] =& \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} \\
=& np \sum_{k=1}^{n} k \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
=& np \sum_{k=1}^{n} (k-1+1) \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
=& n(n-1)p^2 \sum_{k=2}^{n} \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k} \\
&+ np \sum_{k=1}^{n} (k-1+1) \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \\
=& n(n-1)p^2 + np
\end{aligned} \tag{4.4.8}$$

$$\begin{aligned}
\operatorname{var}(X) &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 \\
&= n(n-1)p^2 + np - n^2 p^2 \\
&= np(1-p)
\end{aligned} \tag{4.4.9}$$

Let $n = 1$, then the binomial distribution is the Bernoulli distribution.

Example 4.6 (Expectation and Variance of Poisson Distribution)

$$\begin{aligned}
\mathbf{E}[X] &= \sum_{k=0}^{\infty} k\mathrm{e}^{-\lambda}\frac{\lambda^k}{k!} \\
&= \lambda\mathrm{e}^{-\lambda}\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda} \\
&= \lambda
\end{aligned} \tag{4.4.10}$$

$$\begin{aligned}
\mathbf{E}[X^2] &= \sum_{k=0}^{\infty} k^2\mathrm{e}^{-\lambda}\frac{\lambda^k}{k!} \\
&= \lambda\mathrm{e}^{-\lambda}\sum_{k=1}^{\infty} k\frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda\mathrm{e}^{-\lambda}\sum_{k=1}^{\infty}(k-1+1)\frac{\lambda^{k-1}}{(k-1)!} \\
&= \lambda\mathrm{e}^{-\lambda}\left(\lambda\sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} + \sum_{k=1}^{\infty}\frac{\lambda^{k-1}}{(k-1)!}\right) \\
&= \lambda\mathrm{e}^{-\lambda}(\lambda\mathrm{e}^{\lambda} + \mathrm{e}^{\lambda}) \\
&= \lambda^2 + \lambda
\end{aligned} \tag{4.4.11}$$

$$\begin{aligned}
\mathrm{var}(X) &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 \\
&= \lambda^2 + \lambda - \lambda^2 \\
&= \lambda
\end{aligned} \tag{4.4.12}$$

Chapter 5 Multiple Random Variables

5.1 Conditioning

Definition 5.1 (Conditional PMF)

The conditional PMF of X given A is defined as

$$p_{X|A}(x) = \mathbf{P}(X = x|A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)} \tag{5.1.1}$$

Specifically, if given $Y = y$, then

$$p_{X|Y=y}(x) = \mathbf{P}(X = x|Y = y) = \frac{\mathbf{P}(\{X = x\} \cap \{Y = y\})}{\mathbf{P}(Y = y)} \tag{5.1.2}$$

♣

Since $\mathbf{P}(A) = \sum_x \mathbf{P}(\{X = x\} \cap A)$, we have $\sum_x p_{X|A}(x) = 1$. The conditional PMF limits the sample space to A.

Definition 5.2 (Conditional Expectation)

The conditional expectation of X given A is defined as

$$\mathbf{E}[X|A] = \sum_x x p_{X|A}(x) \tag{5.1.3}$$

Specifically, if given $Y = y$, then

$$\mathbf{E}[X|Y = y] = \sum_x x p_{X|Y}(x|y) \tag{5.1.4}$$

♣

Theorem 5.1 (Total Expectation Theorem)

Partition the sample space into disjoint events $A_1, A_2, \cdots$, then

$$\begin{aligned} \mathbf{P}(B) &= \sum_i \mathbf{P}(A_i)\mathbf{P}(B|A_i) \\ p_X(x) &= \sum_i \mathbf{P}(A_i) p_{X|A_i}(x) \\ \mathbf{E}[X] &= \sum_i \mathbf{P}(A_i)\mathbf{E}[X|A_i] \end{aligned} \tag{5.1.5}$$

From line 2 to line 3, we multiply x on both sides and sum over x.

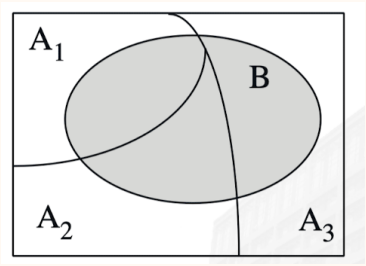


Figure 5.1: Total Expectation Theorem

♡

Example 5.1 (Geometric PMF 1) Toss a fair coin independently until a head occurs. Let X be the number of tosses. What is the PMF, expectation, and variance of X?

Solution

$$\begin{aligned} \mathbf{E}[X] &= \sum_{k=1}^{\infty} k\mathbf{P}(X = k) = \sum_{k=1}^{\infty} k(1-p)^{k-1}p = \frac{1}{p} \\ var(X) &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{1-p}{p^2} \end{aligned} \tag{5.1.6}$$

Property 5.1 (Memoryless Property) *Given that $X > 2$, the random variable $X - 2$ has same geometric PMF with X (not given that $X > 2$).*

$$\begin{aligned} p_{X|X>2}(x) &= \frac{\mathbf{P}(\{X > 2\} \cap \{X = x\})}{\mathbf{P}(X > 2)} \\ &= \frac{(1-p)^{k-1}p}{1-p-p(1-p)} = (1-p)^{k-3}p \end{aligned} \tag{5.1.7}$$

If we shift k by 2, then the PMF is the same. That is to say, the random variable $X - 2$ given $X > 2$ has the same geometric PMF with X, and thus $\mathbf{E}[(X-2)|X>2] = \mathbf{E}[X]$.

Example 5.2 (Geometric PMF 2) Toss a fair coin independently until a head occurs. Let X be the number of tosses. What is the PMF, expectation, and variance of X?

Solution *If we use the memoryless property, then*

$$\begin{aligned} \mathbf{E}[X] &= \mathbf{P}(X=1)\mathbf{E}[X|X=1] + \mathbf{P}(X>1)\mathbf{E}[X|X>1] \\ &= p + (1-p)\mathbf{E}[(X-1+1)|X>1] \\ &= p + (1-p)(1+\mathbf{E}[X]) \\ &\Rightarrow \mathbf{E}[X] = \frac{1}{p} \end{aligned} \tag{5.1.8}$$

Similarly

$$\begin{aligned} \mathbf{E}[X^2] &= \mathbf{P}(X=1)\mathbf{E}[X^2|X=1] + \mathbf{P}(X>1)\mathbf{E}[X^2|X>1] \\ &= p + (1-p)\mathbf{E}[(X-1+1)^2|X>1] \\ &= p + (1-p)(1+2\mathbf{E}[X]+\mathbf{E}[X^2]) \\ &\Rightarrow \mathbf{E}[X^2] = \frac{2-p}{p^2} \end{aligned} \tag{5.1.9}$$

Then

$$var(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{1-p}{p^2} \tag{5.1.10}$$

5.2 Multiple Discrete Random Variables

Definition 5.3 (Joint PMF)

The joint PMF of X and Y is defined as

$$p_{X,Y}(x,y) = \mathbf{P}(\{X=x\} \cap \{Y=y\}) \tag{5.2.1}$$

♣

Property 5.2 (Properties of Joint PMF)

- *Added up to 1:* $\sum_x \sum_y p_{X,Y}(x,y) = 1$.
- *Marginal PMF:* $p_X(x) = \sum_y p_{X,Y}(x,y)$, $p_Y(y) = \sum_x p_{X,Y}(x,y)$. *(Total Probability Theorem)*
- *Conditional PMF:* $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$, $p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$.

If we have a RV function of multiple RVs: $Z = g(X,Y)$, then the PMF of Z is

$$p_Z(z) = \sum_{x,y:g(x,y)=z} p_{X,Y}(x,y) \tag{5.2.2}$$

Definition 5.4 (Expectation of Multiple Random Variables)

Recall that

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$$

Then the expectation of $g(X,Y)$ is

$$\mathbf{E}[g(X,Y)] = \sum_{x,y} g(x,y)p_{X,Y}(x,y) \tag{5.2.3}$$

♣

Property 5.3 (Properties of Expectation)

- *In general,* $\mathbf{E}[X+Y] \neq \mathbf{E}[X] + \mathbf{E}[Y]$.
- $\mathbf{E}[\alpha X + \beta Y + \gamma] = \alpha\mathbf{E}[X] + \beta\mathbf{E}[Y] + \gamma$. ***This property holds for any RVs, despite of the independence.*** *(go back to the definition of expectation and you will see the linearity, which is also the reason of property 1.)*

Example 5.3 (Coin tossing) Toss a coin independently. The coin may not be fair.

Solution

$$X_i = \begin{cases} 1, & \textit{if the } i\textit{-th toss is head} \\ 0, & \textit{otherwise} \end{cases} \tag{5.2.4}$$

The sum $X = \sum_{i=1}^{n} X_i$. *Then*

$$\mathbf{E}[X] = \sum_{i=1}^{n} \mathbf{E}[X_i] = np \tag{5.2.5}$$

$$\begin{aligned} var(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2 &= (\sum_{i=1}^{N} X_i^2 + \sum_{i \neq j} X_i X_j) - n^2p^2 \\ &= np + n(n-1)p^2 - n^2p^2 = np(1-p) \end{aligned} \tag{5.2.6}$$

5.3 Independence

Definition 5.5 (Definition of Independence)

Two RVs X *and* Y *are independent if*

$$p_{X,Y}(x,y) = p_X(x) \cdot p_Y(y) \tag{5.3.1}$$

for all x *and* y.

♣

Property 5.4 (Properties of Independence)

- $\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)] \cdot \mathbf{E}[h(Y)]$. *(go back to the definition of expectation, divide the sum into two parts, and you will see the independence.)*
- $var(g(X) + h(Y)) = var(g(X)) + var(h(Y))$. *(use* $var(A+B) = \mathbf{E}[(A+B)^2] - \mathbf{E}[A+B]^2$.*)*

Definition 5.6 (Definition of Conditional Independence)

Two RVs X *and* Y *are conditionally independent given* A *if*

$$p_{X,Y|A}(x,y) = p_{X|A}(x) p_{Y|A}(y) \tag{5.3.2}$$

for all x *and* y.

♣

Example 5.4 (Data packet problem) A network system with n nodes randomly redistributes each node's data packet through a central processor. Let X be the number of nodes that receive their original data packets. Find $\mathbf{E}[X]$ and var(X).

Solution *Define* X_i *as the indicator of the* i*-th node. Then* $X = \sum_{i=1}^{n} X_i$.

$$\mathbf{E}[X] = \sum_{i=1}^{n} \mathbf{E}[X_i] = n \cdot \frac{1}{n} = 1 \tag{5.3.3}$$

$$\begin{aligned} var(X) = \mathbf{E}[X^2] - \mathbf{E}[X]^2 &= \sum_{i=1}^{n} \mathbf{E}[X_i^2] + \sum_{i \neq j} \mathbf{E}[X_i X_j] - 1 \\ &= n \cdot \frac{1}{n} + \sum_{i=1}^{n} (\mathbf{P}(X_i = 1) \cdot \mathbf{P}(X_j = 1 | X_i = 1)) - 1 \\ &= n(n-1) \cdot \frac{1}{n(n-1)} = 1 \end{aligned} \tag{5.3.4}$$

Chapter 6 Continuous Random Variables

6.1 Continuous RVs and PDFs

Definition 6.1 (Continuous Random Variables)

A RV X is continuous if there exists a nonnegative function $f_X(x)$, called ***probability density function****, such that*

$$\mathbf{P}(X \in A) = \int_A f_X(x)\,\mathrm{d}x \tag{6.1.1}$$

Especially, if $A \subseteq \mathcal{B}(\mathbb{R})$, then

$$\mathbf{P}(a \le X \le b) = \int_a^b f_X(x)\,\mathrm{d}x \tag{6.1.2}$$

♣

Property 6.1 (Properties of PDF) *Let X be a continuous RV with PDF $f_X(x)$, then*

- $f_X(x) \ge 0$, *and $f_X(x)$ can be any nonnegative number. (even $+\infty$)*
- $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$.
- $\mathbf{P}(x \le X \le x+\delta) = \int_x^{x+\delta} f_X(x)\,\mathrm{d}x \approx f_X(x)\cdot\delta$.

6.2 Expectation and Variance

Definition 6.2 (Definitions of Expectation and Variance)

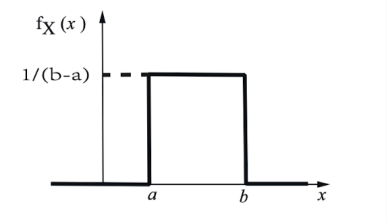
Let X be a continuous RV with PDF $f_X(x)$, then

$$\begin{aligned} \mathbf{E}[X] &= \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x \\ \mathbf{E}[g(X)] &= \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x \end{aligned} \tag{6.2.1}$$

$$\begin{aligned} var(X) = \mathbf{E}[(X-\mathbf{E}[X])^2] &= \int_{-\infty}^{\infty} (x-\mathbf{E}[X])^2 f_X(x)\,\mathrm{d}x \\ &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 \end{aligned} \tag{6.2.2}$$

♣

Example 6.1 (Uniform RV) Consider a RV X that takes value in an interval $[a,b]$ with PDF



$$\mathbf{E}[X] = \int_a^b \frac{x}{b-a}\,\mathrm{d}x = \frac{a+b}{2} \tag{6.2.3}$$

$$\mathrm{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12} \tag{6.2.4}$$

Example 6.2 (Exponential RV) Consider an exponential RV X that has a PDF of the form

$$f_X(x) = \lambda \mathrm{e}^{-\lambda x},\ x \ge 0 \tag{6.2.5}$$

$$\mathbf{E}[X] = \int_0^{\infty} x\lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = (-x\mathrm{e}^{-\lambda x})\Big|_0^{\infty} + \int_0^{\infty} \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \frac{1}{\lambda} \tag{6.2.6}$$

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = (-x^2 \mathrm{e}^{-\lambda x})\Big|_0^\infty + \int_0^\infty 2x\mathrm{e}^{-\lambda x}\,\mathrm{d}x - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \tag{6.2.7}$$

6.3 Comulative Distribution Function

We want to describe both discrete and continuous RVs in a unified way. An intuition is to "accumulate" the probability "up to" the value x.

Definition 6.3 (CDF)

The cumulative distribution function (CDF) of a RV X is defined as

$$F_X(x) = \mathbf{P}(X \le x) = \begin{cases} \sum_{t \le x} p_X(t), & \textit{discrete} \\ \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t\,, & \textit{continuous} \end{cases} \tag{6.3.1}$$

♣

Example 6.3 (Some examples of CDF)

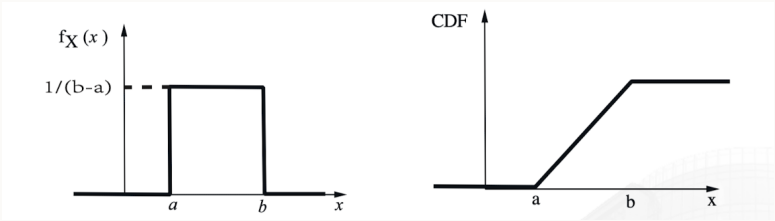


Figure 6.1: CDF: Continuous RV

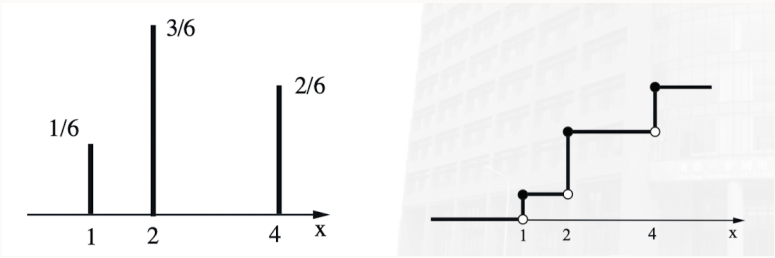


Figure 6.2: CDF: Discrete RV

Property 6.2 (Properties of CDF)

- *Monotonically nondecreasing:* $x \le y \Rightarrow F_X(x) \le F_X(y)$. *(since* $p_X(x) \ge 0$ *and* $f_X(x) \ge 0$*)*
- $F_X(x) \to 0$ *as* $x \to -\infty$ *and* $F_X(x) \to 1$ *as* $x \to \infty$.
- *When X is continuous:* $F_X(x)$ *is a continuous function,* $f_X(x) = \frac{\mathrm{d}F_X}{\mathrm{d}x}$.
- *When X is discrete:* $F_X(x)$ *i piecewise constant function,* $p_X(k) = F_X(k) - F_X(k-1)$.

Example 6.4 (CDF of Geometry RV) Let X be a geometry RV with parameter p, then

$$\begin{aligned} &\mathbf{P}(X = k) = p(1-p)^{k-1} \\ \Rightarrow &F_X(x) = \sum_{k \le x} p(1-p)^{k-1} = 1 - (1-p)^{\lfloor x \rfloor} \end{aligned} \tag{6.3.2}$$

Example 6.5 (Maximum test score) Assume that each test takes one of the values from 1 to 10 with equal probability 1/10 independently. What is the PMF of the maximum of three test scores?

$$X = \max\{X_1, X_2, X_3\} \tag{6.3.3}$$

Solution *We have the CDF of X as*

$$\begin{aligned} F_X(x) &= \mathbf{P}(X \le x) \\ &= \mathbf{P}(\max\{X_1, X_2, X_3\} \le x) \\ &= \mathbf{P}(X_1 \le x, X_2 \le x, X_3 \le x) \\ &= \mathbf{P}(X_1 \le x) \cdot \mathbf{P}(X_2 \le x) \cdot \mathbf{P}(X_3 \le x) \\ &= F_{X_1}(x) \cdot F_{X_2}(x) \cdot F_{X_3}(x) \\ &= \left(\frac{\lfloor x \rfloor}{10}\right)^3 \end{aligned} \tag{6.3.4}$$

and the PMF of X is

$$p_X(x) = F_X(x) - F_X(x-1) = \left(\frac{\lfloor x \rfloor}{10}\right)^3 - \left(\frac{\lfloor x \rfloor - 1}{10}\right)^3 \tag{6.3.5}$$

Similarly, this approach can be introduced to the minimum of three test scores (need to reverse by $1 - \mathbf{P}(xxx)$).

6.4 Normal and Exponential RVs

Definition 6.4 (Gaussian (Normal) RV)

- *Standard normal $N(0,1)$:*

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-x^2/2} \tag{6.4.1}$$

  which has $\mathbf{E}[X] = 0$ and $\mathrm{var}(X) = 1$. (intergrate by part)
- *General normal $N(\mu, \sigma^2)$:*

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-(x-\mu)^2/2\sigma^2} \tag{6.4.2}$$

  which has $\mathbf{E}[X] = \mu$ and $\mathrm{var}(X) = \sigma^2$. (let $Y = (X-\mu)/\sigma$ and Y is standard normal)
- *The CDF of standard normal is hard to calculate, we denote it as*

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \mathrm{e}^{-t^2/2} \, \mathrm{d}t \tag{6.4.3}$$

  If $X \sim N(\mu, \sigma^2)$, then $\mathbf{P}(X \le x) = \mathbf{P}(\frac{X-\mu}{\sigma} \le \frac{x-\mu}{\sigma}) = \Phi(\frac{x-\mu}{\sigma})$.

♣

Definition 6.5 (Exponential RV)

A RV X is exponential with parameter λ if it has PDF

$$f_X(x) = \lambda \mathrm{e}^{-\lambda x}, \ x \ge 0 \tag{6.4.4}$$

which has $\mathbf{E}[X] = 1/\lambda$ and $\mathrm{var}(X) = 1/\lambda^2$, and the CDF is

$$F_X(x) = 1 - \int_x^{+\infty} \lambda \mathrm{e}^{-\lambda t} \, \mathrm{d}t = 1 - \mathrm{e}^{-\lambda x} \tag{6.4.5}$$

♣

A continuous exponential RV also has the memoryless property as discrete exponential RV, which means

Property 6.3 (Memoryless Property) *Let X be an exponential RV with parameter λ, then for all $x, c \ge 0$,*

$$\mathbf{P}(X > x + c \mid X > c) = \mathbf{P}(X > x) = \mathrm{e}^{-\lambda x} \tag{6.4.6}$$

Because $\mathrm{e}^{-\lambda x} \sim x + 1, \ x \to 0$, we have

Property 6.4 (Continuous-time analog of the geometric)

- $\mathbf{P}(0 \le X \le \delta) = 1 - \mathrm{e}^{-\lambda\delta} \approx \lambda\delta$.
- $\mathbf{P}(c \le X \le c + \delta \mid X > c) \approx \lambda\delta$.

Chapter 7 Multiple Continuous Random Variables

7.1 Multiple Continuous Random Variables

Definition 7.1 (Joint PDF)

The two continuous RVs X and Y, with the same experiment, are jointly continuous if they can be described by a joint PDF $f_{X,Y}(x,y)$, where $f_{X,Y}(x,y) \geq 0$ and satisfies

$$\mathbf{P}((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \tag{7.1.1}$$

Especially, if $A \subseteq \mathcal{B}(\mathbb{R}^2)$, then

$$\mathbf{P}((X,Y) \in [a,b] \times [c,d]) = \int_c^d \int_a^b f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \tag{7.1.2}$$

♣

Property 7.1 (Properties of Joint PDFs)

- *Normalization:* $\iint_\Omega f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = 1$.
- *Interpretation:* $f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y$ *is the probability that* (X,Y) *lies in the small rectangle* $[x, x+\mathrm{d}x] \times [y, y+\mathrm{d}y]$.
- *Marginal PDFs:* $f_X(x) = \int_{\Omega_Y} f_{X,Y}(x,y)\,\mathrm{d}y$ *and* $f_Y(y) = \int_{\Omega_X} f_{X,Y}(x,y)\,\mathrm{d}x$.

Definition 7.2 (Joint CDFs)

If X and Y are two continuous RVs associated with the same experiment, then the joint CDF of X and Y is defined as

$$F_{X,Y}(x,y) = \mathbf{P}(X \leq x, Y \leq y) \tag{7.1.3}$$

If X and Y can be described by a joint PDF $f_{X,Y}(x,y)$, then

$$F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(t,s)\,\mathrm{d}t\,\mathrm{d}s \tag{7.1.4}$$

Conversely, if $F_{X,Y}(x,y)$ is differentiable, then

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} \tag{7.1.5}$$

♣

Definition 7.3 (Expectation and Variance)

Let X and Y be two continuous RVs with joint PDF $f_{X,Y}(x,y)$, then

$$\begin{aligned} &\mathbf{E}[g(X,Y)] = \iint_\Omega g(x,y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &var(g(X,Y)) = \mathbf{E}[g(X,Y) - \mathbf{E}[g(X,Y)]]^2 = \mathbf{E}[g(X,Y)^2] - \mathbf{E}[g(X,Y)]^2 \end{aligned} \tag{7.1.6}$$

♣

Definition 7.4 (Independence)

X and Y are independent if their joint PDF can be factorized as

$$f_{X,Y}(x,y) = f_X(x) f_Y(y),\ \forall x, y \tag{7.1.7}$$

♣

Example 7.1 (Buffon's needle) Parallel lines are at distance d, throw a needle of length l (assume $l < d$). Find $\mathbf{P}$(needle intersects one of the lines).

Solution *Let X be the distance from the center of the needle to the nearest line, and Θ be the angle between the needle and the lines, where X and Θ are independent. Given the symmetry characteristics, it is well to assume X is uniform in $[0, d/2]$ and Θ is uniform in $[0, \pi/2]$. Then the joint PDF is*

$$f_{X,\Theta}(x,\theta) = \frac{2}{d} \cdot \frac{2}{\pi}, \quad 0 \leq x \leq \frac{d}{2}, 0 \leq \theta \leq \frac{\pi}{2} \tag{7.1.8}$$

The needle intersects if $X \le l/2 \sin\Theta$, and the probability is

$$\begin{aligned}\mathbf{P}(X \le \frac{l}{2}\sin\Theta) &= \iint_{X \le \frac{l}{2}\sin\Theta} f_{X,\Theta}(x,\theta)\,\mathrm{d}x\,\mathrm{d}\theta \\ &= \frac{4}{\pi d}\int_0^{\pi/2}\int_0^{(l/2)\sin\theta} \mathrm{d}x\,\mathrm{d}\theta = \frac{2l}{\pi d}\end{aligned} \tag{7.1.9}$$

7.2 Conditioning and Independence

Definition 7.5 (Definition of Conditional PDFs)

Let X and Y be two continuous RVs with joint PDF $f_{X,Y}(x,y)$, then the conditional PDF of X given $Y = y$ is defined as

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \tag{7.2.1}$$

♣

Remark There's a paradox that the denominator $\mathbf{P}(Y = y) = 0$ for continuous RVs, but we can let Y be a little fatter: $\mathbf{P}(y \le Y \le y + \delta) \approx f_Y(y) \cdot \delta$ and let $\delta \to 0$ after we do division.

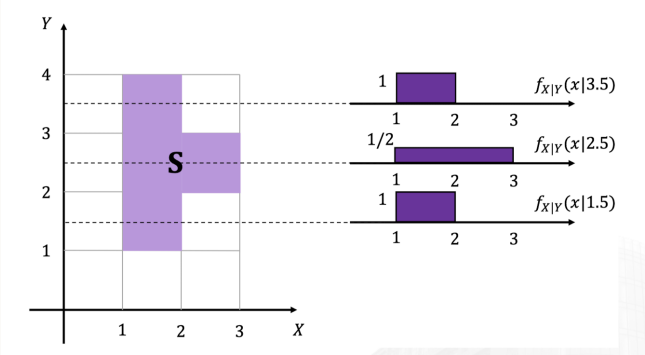


Figure 7.1: Visualization of Conditional PDF

Example 7.2 (Stick-breaking) Break a stick of length l twice: break at X: uniform in $[0,l]$; break again at Y: uniform in $[0,X]$. Find $\mathbf{E}[Y]$.

Solution

$$f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y \mid x) = \frac{1}{l}\cdot\frac{1}{x}, \quad 0 \le y \le x \le l \tag{7.2.2}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)\,\mathrm{d}x = \int_y^l \frac{1}{l}\cdot\frac{1}{x}\,\mathrm{d}x = \frac{1}{l}\ln\left(\frac{l}{y}\right), \quad 0 \le y \le l \tag{7.2.3}$$

$$\mathbf{E}[Y] = \int_{-\infty}^{+\infty} y f_Y(y)\,\mathrm{d}y = \int_0^l y\cdot\frac{1}{l}\ln\left(\frac{l}{y}\right)\mathrm{d}y = \frac{l}{4} \tag{7.2.4}$$

Definition 7.6 (Conditional Expectation)

The conditional expectation of X given that $Y = y$ has happened is defined by

$$\mathbf{E}[X \mid Y = y] = \int_{-\infty}^{+\infty} x f_{X|Y}(x \mid y)\,\mathrm{d}x \tag{7.2.5}$$

For a function $g(X,Y)$, the conditional expectation is

$$\mathbf{E}[g(X,Y) \mid Y = y] = \int_{-\infty}^{+\infty} g(x,y) f_{X|Y}(x \mid y)\,\mathrm{d}x \tag{7.2.6}$$

♣

Remember how we prove the total expectation theorem in discrete case: we multiple x on both side of the equation of conditional probability and sum over x. Now we can do the same thing in continuous case.

Theorem 7.1 (Total Expectation Theorem)

Let X and Y be two continuous RVs, then

$$\mathbf{E}[X] = \mathbf{E}_Y\left[\mathbf{E}_{X|Y}[X \mid Y]\right] = \int_{-\infty}^{+\infty} \mathbf{E}[X \mid Y = y] f_Y(y)\,\mathrm{d}y \tag{7.2.7}$$

♡

Solution ***Another way to solve Example 7.2.***

Conditioning on $X = x$, we have

$$\mathbf{E}[Y \mid X = x] = \int_0^x y f_{Y|X}(y \mid x)\,\mathrm{d}y = \int_0^x y \cdot \frac{1}{x}\,\mathrm{d}y = \frac{x}{2} \tag{7.2.8}$$

With total expectation theorem, we have

$$\mathbf{E}[Y] = \mathbf{E}_X\left[\mathbf{E}[Y \mid X]\right] = \int_0^l \frac{x}{2} f_X(x)\,\mathrm{d}x = \int_0^l \frac{x}{2l}\,\mathrm{d}x = \frac{l}{4} \tag{7.2.9}$$

Definition 7.7 (Independence)

Two continuous RVs X and Y are independent if their joint PDF can be factorized as

$$f_{X,Y}(x, y) = f_X(x) f_Y(y),\ \forall x, y \tag{7.2.10}$$

♣

Similar to discrete case, the following properties hold:

Property 7.2 (Properties of Independence)

- $\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$ *if X and Y are independent.*
- $var(X + Y) = var(X) + var(Y)$ *if X and Y are independent.*

7.3 Bayes' Theorem

Change a little bit from $f_{X,Y}(x, y) = f_{X|Y}(x \mid y) f_Y(y)$, we have

Theorem 7.2 (Bayes' Theorem)

Let X and Y be two continuous RVs, then

$$f_{X|Y}(x \mid y) = \frac{f_{Y|X}(y \mid x) f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y \mid x) f_X(x)}{\int_{-\infty}^{+\infty} f_{Y|X}(y \mid t) f_X(t)\,\mathrm{d}t} \tag{7.3.1}$$

which can be interpreted using $f(x)\,\mathrm{d}x = \mathbf{P}(x \le X \le x + \mathrm{d}x)$.

♡

Chapter 8 Derived Distributions and Entropy

8.1 Derived Distributions

We've already known the Distribution of a RV X and we want to find the distribution of $Y = g(X)$. In discrete case we can calculate the PMF for each possible value of Y like $\mathbf{P}(Y = y) = \sum_{x:g(x)=y} \mathbf{P}(X = x)$. For continuous case we cannot simply do such thing because the conservation of PDF is not guaranteed under summation. Instead, CDF is standardized and we can use it to find the PDF of Y.

Theorem 8.1 (Principal Method for Derived Distribution)

Two-step method to find the distribution of $Y = g(X)$:

1. *Find the CDF of Y:*
$$F_Y(y) = \mathbf{P}(Y \le y) = \mathbf{P}(g(X) \le y) \tag{8.1.1}$$
2. *Differentiate the CDF to find the PDF:*
$$f_Y(y) = \frac{\mathrm{d}F_Y}{\mathrm{d}y} \tag{8.1.2}$$

♡

Example 8.1 (Taking a Train) A professor is taking the high-speed railway from Beijing to Shanghai. Suppose that the speed of the train is uniformly distributed between 280 km/h and 350 km/h. The distance between the two cities is 1400 km. What is the distribution of the duration of the trip?

Solution *Let $T(V) = \frac{1400}{V}$ be the duration of the trip. The speed V is uniformly distributed in $[280, 350]$, so the CDF of V is*

$$F_V(v) = \begin{cases} 0, & v < 280 \\ \frac{v-280}{350-280}, & 280 \le v < 350 \\ 1, & v \ge 350 \end{cases} \tag{8.1.3}$$

The CDF of T is

$$F_T(t) = \mathbf{P}(T \le t) = \mathbf{P}\left(\frac{1400}{V} \le t\right) = \mathbf{P}\left(V \ge \frac{1400}{t}\right) = 1 - F_V\left(\frac{1400}{t}\right) \tag{8.1.4}$$

$$F_T(t) = \begin{cases} 0, & t < 4 \\ 5 - \frac{20}{t}, & 4 \le t < 5 \\ 1, & t \ge 5 \end{cases} \tag{8.1.5}$$

The PDF of T is

$$f_T(t) = \frac{\mathrm{d}F_T}{\mathrm{d}t} = \begin{cases} 0, & t < 4 \\ \frac{20}{t^2}, & 4 \le t < 5 \\ 0, & t \ge 5 \end{cases} \tag{8.1.6}$$

Example 8.2 (PDF of $Y = aX + b$) Let X be a RV with PDF $f_X(x)$, and $Y = aX + b$ where $a \neq 0$, then the PDF of Y is

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \tag{8.1.7}$$

Solution *The CDF of Y is*

$$F_Y(y) = \mathbf{P}(Y \le y) = \mathbf{P}(aX + b \le y) = \mathbf{P}\left(X \le \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right) \tag{8.1.8}$$

And then differentiating the CDF using chain rule

$$f_Y(y) = \frac{\mathrm{d}F_Y}{\mathrm{d}y} = \frac{1}{a}\frac{\mathrm{d}F_X}{\mathrm{d}\frac{y-b}{a}} = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) \tag{8.1.9}$$

If $a < 0$, the inequality will be reversed, and a minus sign will be added to the derivative. So we have

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right) \tag{8.1.10}$$

Example 8.3 (CDF and PDF of Normal Distribution) The PDF of standard normal distribution is

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{x^2}{2}} \tag{8.1.11}$$

whose CDF cannot be expressed in closed form and is defined as

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \mathrm{e}^{-\frac{t^2}{2}} \,\mathrm{d}t \tag{8.1.12}$$

Now, a RV Y is normally distributed with parameters μ and σ^2. Let $Z = \frac{Y-\mu}{\sigma}$, then Z is standard normal distributed. The CDF of Y is

$$F_Y(y) = \mathbf{P}(Y \le y) = \mathbf{P}\left(\frac{Y-\mu}{\sigma} \le \frac{y-\mu}{\sigma}\right) = \mathbf{P}(Z \le \frac{y-\mu}{\sigma}) = \Phi\left(\frac{y-\mu}{\sigma}\right) \tag{8.1.13}$$

And the PDF of Y is

$$f_Y(y) = \frac{\mathrm{d}F_Y}{\mathrm{d}y} = \frac{\mathrm{d}\Phi\left(\frac{y-\mu}{\sigma}\right)}{\mathrm{d}y} = \frac{1}{\sigma}\Phi'\left(\frac{y-\mu}{\sigma}\right) = \frac{1}{\sigma} f_Z\left(\frac{y-\mu}{\sigma}\right) = \frac{1}{\sqrt{2\pi}\sigma} \mathrm{e}^{-\frac{(y-\mu)^2}{2\sigma^2}} \tag{8.1.14}$$

For monotonic functions $g(x)$

Theorem 8.2 (PDF of a Strictly Monotonic Function)

Suppose that $g(\cdot)$ is strictly monotonic, then g has an inverse $h(\cdot) \triangleq g^{-1}(\cdot)$. Assume h is differentiable and the PDF of Y is

$$f_Y(y) = f_X(h(y))\left|\frac{\mathrm{d}h}{\mathrm{d}y}\right| \tag{8.1.15}$$

♡

Proof

$$F_Y(y) = \mathbf{P}(Y \le y) = \mathbf{P}(g(X) \le y) = \mathbf{P}(X \le h(y)) = F_X(h(y)) \tag{8.1.16}$$

$$f_Y(y) = \frac{\mathrm{d}F_Y}{\mathrm{d}y} = f_X(h(y))\frac{\mathrm{d}h}{\mathrm{d}y} \tag{8.1.17}$$

If g is strictly decreasing, the inequality will be reversed, and a minus sign will be added to the derivative. So we have

$$f_Y(y) = f_X(h(y))\left|\frac{\mathrm{d}h}{\mathrm{d}y}\right| \tag{8.1.18}$$

Remark If g is a multiple-to-multiple function, $\left|\frac{\mathrm{d}h}{\mathrm{d}y}\right|$ becomes Jacobian determinant.

8.2 Entropy

Definition 8.1 (Entropy)

The entropy of a discrete RV X with PMF $p_X(x)$ is defined as

$$H(X) = -\sum_x p_X(x) \ln p_X(x) \tag{8.2.1}$$

The entropy of a continuous RV X with PDF $f_X(x)$ is defined as

$$H(X) = -\int_{-\infty}^{+\infty} f_X(x) \ln f_X(x) \,\mathrm{d}x \tag{8.2.2}$$

♣

Remark

- The entropy of a discrete RV is non-negative, while the entropy of a continuous RV can be negative (if $f_X(x) > 1$).
- The entropy describes the uncertainty of a random experiment.
- The base of the logarithm can be different, which is equal to multiplying by a constant.
  - If the base is 2, the unit is bit.

- If the base is e, the unit is nat.

Remark

Why is entropy in the form of the expectation of $-\ln f_X(x)$?

The entropy of a RV is defined as the expectation of the self-information $H(X) = \mathbf{E}[I(p(X))]$. According to Shannon, self-information shoudle satisfies the following axioms:

- Self-information is and only is a function of probability $I(p)$.
- If probability $p = p_1 p_2$, then $I(p) = I(p_1) + I(p_2)$.
- $I(p)$ is continuous.

The only function that satisfies these axioms is $I(p) = k \ln p$. We add a negative sign to make it positive, and the constant k is equivalent to the base of the logarithm.

Example 8.4 (Entropy of Bernoulli RV) Let X be a Bernoulli RV with $\mathbf{P}(X = 1) = p$, $\mathbf{P}(X = 0) = 1 - p$. Find the entropy of X.

Solution *THe entropy of X in bits is*

$$H(X) = -p \ln p - (1 - p) \ln(1 - p) \tag{8.2.3}$$

which is maximized when $p = 0.5$ (the most uncertain case), and is minimized when $p = 0$ or $p = 1$ (the most certain case).

8.3 Maximum Entropy Distributions

Optional, to be completed.

Chapter 9 Convolution, Covariance, Correlation, and Conditional Expectation

9.1 Convolution

Definition 9.1 (Convolution)

The convolution of two functions f and g is defined as

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(x')g(x - x')\,\mathrm{d}x' \tag{9.1.1}$$

The convolution of two discrete functions is defined as

$$(f * g)(k) = \sum_{k'=-\infty}^{+\infty} f(k')g(k - k') \tag{9.1.2}$$

♣

When considering a RV $W = X + Y$ where X and Y are independent discrete RVs

$$p_W(w) = \mathbf{P}(X + Y = w) = \sum_{\{(x,y):x+y=w\}} p_X(x)p_Y(y) = \sum_x p_X(x)p_Y(w - x) \tag{9.1.3}$$

For continuous cases

$$f_W(w) = \mathbf{P}(X + Y = w) = \int_{-\infty}^{+\infty} f_X(x)f_Y(w - x)\,\mathrm{d}x \tag{9.1.4}$$

If X and Y are dependent, let $f_{X,Y}(x, y)$ be the joint PDF of X and Y, then

$$f_W(w) = \int_{-\infty}^{+\infty} f_X(x)f_{X,Y}(x, w - x)\,\mathrm{d}x \tag{9.1.5}$$

Remark This is somehow similar to LTI systems in SS. When the system is not time-invariant, $h(t - \tau)$ becomes $h(t, \tau)$.

Remark

- Conditional CDF: $F_{W|X}(w \mid x) = F_Y(w - x)$
- Conditional PDF: $f_{W|X}(w \mid x) = f_Y(w - x)$
- Joint PDF: $f_{W,X}(w, x) = f_X(x)f_Y(w - x)$
- Marginal PDF: $f_W(w) = \int_{-\infty}^{+\infty} f_{W,X}(w, x)\,\mathrm{d}x = \int_{-\infty}^{+\infty} f_X(x)f_Y(w - x)\,\mathrm{d}x$

Example 9.1 (Convolution of Uniform RVs) The RVs X and Y are independent and uniformly distributed in the interval $[0, 1]$. Find the PDF of $Z = X + Y$.

Solution

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(z - x)\,\mathrm{d}x = \begin{cases} 0, & z < 0 \\ z, & 0 \le z < 1 \\ 2 - z, & 1 \le z < 2 \\ 0, & z \ge 2 \end{cases} \tag{9.1.6}$$

Example 9.2 (Sum of Two Independent Normal RVs) $X \sim N(\mu_x, \sigma_x^2), Y \sim N(\mu_y, \sigma_y^2)$ are independent.

- The joint PDF is

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \mathrm{e}^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} \mathrm{e}^{-\frac{(y-\mu_y)^2}{2\sigma_y^2}} \tag{9.1.7}$$

  which is a constant (as shown in Figure 9.1) on the ellipse where

$$\frac{(x - \mu_x)^2}{2\sigma_x^2} + \frac{(y - \mu_y)^2}{2\sigma_y^2} = const. \tag{9.1.8}$$

- Let $W = X + Y$, the PDF of W is

$$f_W(w) = \int_{-\infty}^{+\infty} f_{X,Y}(x, w - x)\,\mathrm{d}x = \frac{1}{2\pi\sigma_x\sigma_y} \int_{-\infty}^{+\infty} \exp\left(-\frac{(x - \mu_x)^2}{2\sigma_x^2} - \frac{(w - x - \mu_y)^2}{2\sigma_y^2}\right) \mathrm{d}x = \cdots \tag{9.1.9}$$

which is also a normal RV with mean $\mu_w = \mu_x + \mu_y$ and variance $\sigma_w^2 = \sigma_x^2 + \sigma_y^2$.

Proof Obvious after Fourier transformation.

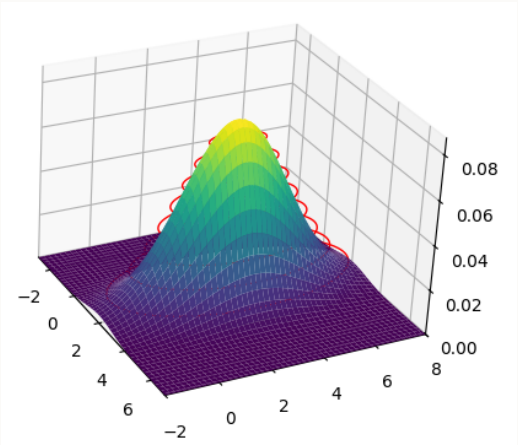


Figure 9.1: The joint PDF of two independent normal RVs.

Example 9.3 (The Difference of Two Independent Exponential RVs) Consider the case where X and Y are independent exponential RVs with parameter λ and μ, respectively. Find the PDF of $Z = X - Y$.

Solution

$$f_Z(z) = f_{X-Y}(z) = \int_{-\infty}^{+\infty} f_X(x) f_{-Y}(z-x)\,\mathrm{d}x = \int_{-\infty}^{+\infty} f_X(x) f_Y(x-z)\,\mathrm{d}x \tag{9.1.10}$$

For $z \geq 0$, $f_X(x)$ is nonzero when $x \geq 0$ and $f_Y(x-z)$ is nonzero when $x \geq z$.

$$f_Z(z) = \int_z^{+\infty} \lambda \mathrm{e}^{-\lambda x} \mu \mathrm{e}^{-\mu(x-z)}\,\mathrm{d}x = \int_z^{+\infty} \lambda\mu \mathrm{e}^{-(\lambda+\mu)x+\mu z}\,\mathrm{d}x = \frac{\lambda\mu}{\lambda+\mu}\mathrm{e}^{-\lambda z} \quad (z \geq 0) \tag{9.1.11}$$

For $z < 0$, $f_X(x)$ is nonzero when $x \geq 0$ and $f_Y(x-z)$ is nonzero when $x \leq z$.

$$f_Z(z) = \int_0^{z} \lambda \mathrm{e}^{-\lambda x} \mu \mathrm{e}^{-\mu(x-z)}\,\mathrm{d}x = \int_0^{z} \lambda\mu \mathrm{e}^{-(\lambda+\mu)x+\mu z}\,\mathrm{d}x = \frac{\lambda\mu}{\lambda+\mu}\mathrm{e}^{\mu z} \quad (z < 0) \tag{9.1.12}$$

9.2 Covariance and Correlation

Definition 9.2 (Covariance)

The covariance of two RVs X and Y is defined as

$$\mathrm{cov}(X,Y) = \mathbf{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbf{E}[XY] - \mu_X\mu_Y \tag{9.2.1}$$

where $\mu_X = \mathbf{E}[X]$ and $\mu_Y = \mathbf{E}[Y]$.

The covariance describes the degree to which two RVs vary together. If X and Y are independent, then $\mathrm{cov}(X,Y) = 0$.

♣

Proof The second equality is because

$$\mathbf{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbf{E}[XY] - \mu_X\mathbf{E}[Y] - \mu_Y\mathbf{E}[X] + \mu_X\mu_Y = \mathbf{E}[XY] - \mu_X\mu_Y \tag{9.2.2}$$

Property 9.1 (Properties of Covariance)

- $\mathrm{cov}(X,X) = \mathrm{var}(X)$
- $\mathrm{cov}(X, aY+b) = a\mathrm{cov}(X,Y)$ *(linearity of expectation)*
- $\mathrm{cov}(X, Y+Z) = \mathrm{cov}(X,Y) + \mathrm{cov}(X,Z)$ *(linearity of expectation)*
- *independent* $\Rightarrow \mathrm{cov}(X,Y) = 0$ *(but not the other way around)*

Theorem 9.1 (Variance of the Sum of Multiple RVs)

$$\operatorname{var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \operatorname{var}(X_i) + \sum_{\{(i,j):j\neq i\}}^{n} \operatorname{cov}(X_i, X_j) \tag{9.2.3}$$

which is somehow similar to the complete square of the quadratic form. ♡

Proof Denote $\widetilde{X_i} = X_i - \mathbf{E}[X_i]$, then

$$\mathbf{E}\left[\widetilde{X_i}^2\right] = \mathbf{E}\left[X_i^2 - 2X_i\mathbf{E}[X_i] + \mathbf{E}[X_i]^2\right] = \mathbf{E}[X_i^2] - 2\mathbf{E}[X_i]\mathbf{E}[X_i] + \mathbf{E}[X_i]^2 = \operatorname{var}(X_i) \tag{9.2.4}$$

$$\mathbf{E}\left[\widetilde{X_i}\widetilde{X_j}\right] = \mathbf{E}[X_iX_j - X_i\mathbf{E}[X_j] - \mathbf{E}[X_i]X_j + \mathbf{E}[X_i]\mathbf{E}[X_j]] = \mathbf{E}[X_iX_j] - \mathbf{E}[X_i]\mathbf{E}[X_j] = \operatorname{cov}(X_i, X_j) \tag{9.2.5}$$

Therefore

$$\begin{aligned}\operatorname{var}\left(\sum_{i=1}^{n} X_i\right) &= \mathbf{E}\left[\left(\sum_{i=1}^{n} X_i - \mathbf{E}[X_i]\right)^2\right] = \mathbf{E}\left[\left(\sum_{i=1}^{n} \widetilde{X_i}\right)^2\right] \\ &= \sum_{i=1}^{n} \mathbf{E}\left[\widetilde{X_i}\right] + \sum_{\{(i,j):j\neq i\}}^{n} \mathbf{E}[X_i, X_j] \\ &= \sum_{i=1}^{n} \operatorname{var}(X_i) + \sum_{\{(i,j):j\neq i\}}^{n} \operatorname{cov}(X_i, X_j)\end{aligned} \tag{9.2.6}$$

Example 9.4 (Data Packet Problem) Consider the data packet problem again, where a router randomly distributes n data packets to n nodes. Find the variance of X, the number of nodes where the i-th node receives the i-th data packet.

Solution *Let X_i be the indicator of the event that the i-th node receives the i-th data packet. Then $X = \sum_{i=1}^{n} X_i$. Note that X_i obeys the Bernoulli distribution with parameter $p = 1/n$. Therefore,*

$$\mathbf{E}[X_i] = \frac{1}{n}, \quad \operatorname{var}(X_i) = \frac{1}{n}\left(1 - \frac{1}{n}\right) \tag{9.2.7}$$

For $i \neq j$

$$\begin{aligned}\operatorname{cov}(X_i, X_j) &= \mathbf{E}[X_iX_j] - \mathbf{E}[X_i]\mathbf{E}[X_j] = \mathbf{P}(X_i = 1, X_j = 1) - \frac{1}{n^2} \\ &= \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{1}{n^2(n-1)}\end{aligned} \tag{9.2.8}$$

Therefore

$$\begin{aligned}\operatorname{var}(X) &= \sum_{i=1}^{n} \operatorname{var}(X_i) + \sum_{\{(i,j):j\neq i\}}^{n} \operatorname{cov}(X_i, X_j) \\ &= n \cdot \frac{1}{n}\left(1 - \frac{1}{n}\right) + n(n-1) \cdot \frac{1}{n^2(n-1)} = 1\end{aligned} \tag{9.2.9}$$

Definition 9.3 (Coefficient of Correlation)

The coefficient of correlation of two RVs X and Y is defined as

$$\rho(X, Y) = \frac{\operatorname{cov}(X, Y)}{\sigma_X\sigma_Y} = \mathbf{E}\left[\frac{X - \mu_X}{\sigma_X} \cdot \frac{Y - \mu_Y}{\sigma_Y}\right] \tag{9.2.10}$$

where $\sigma_X = \sqrt{\operatorname{var}(X)}$, $\sigma_Y = \sqrt{\operatorname{var}(Y)}$, $\mu_X = \mathbf{E}[X]$, and $\mu_Y = \mathbf{E}[Y]$. ♣

According to the linearity of covariance, if $X \to 1000X$ (e.g. change the unit of X from g to kg), then $\operatorname{cov}(X, Y) \to 1000\operatorname{cov}(X, Y)$, which is not what we want. So we need to normalize the covariance. Here we choose σ_X and σ_Y as the normalization factors because $\sigma_X \to a\sigma_X$ when $X \to aX + b$, i.e. σ is proportional to the scale and unrelated to the shift, just like the covariance.

Property 9.2 (Properties of Coefficient)

- $|\rho(X, Y)| \leq 1$
- $|\rho(X, Y)| = 1 \Leftrightarrow (X - \mathbf{E}[X]) = c(Y - \mathbf{E}[Y])$ *(linearly related)*

- *X and Y are independent $\Rightarrow \rho(X,Y)=0$ (but not the other way around)*

Example 9.5 (Data Transmission) Consider n independent transmissions of a binary signal where the probability of transmitting a bit 1 is p and a bit 0 is $1-p$. Let X represent the number of 1's transmitted and Y the number of 0's. Find the correlation coefficient between X and Y .

Solution *Since $X+Y=n$ which means X and Y are linearly related, we have $\rho(X,Y)=-1$.*
Or, we have $X+Y=n$ and $\mathbf{E}[X]+\mathbf{E}[Y]=n$, thus

$$\begin{aligned} & X-\mathbf{E}[X]=-(Y-\mathbf{E}[Y]) \\ \Rightarrow\quad & \operatorname{cov}(X,Y)=\mathbf{E}[(X-\mathbf{E}[X])(Y-\mathbf{E}[Y])]=-\operatorname{var}(X)=-\operatorname{var}(Y) \\ \Rightarrow\quad & \rho(X,Y)=\frac{\operatorname{cov}(X,Y)}{\sigma_X\sigma_Y}=-\frac{\operatorname{var}(X)}{\sqrt{\operatorname{var}(X)}\sqrt{\operatorname{var}(Y)}}=-1 \end{aligned} \tag{9.2.11}$$

9.3 Conditional Expectation

Theorem 9.2 (The Conditional Expectation as an Estimator)

Denote the conditional expectation

$$\hat{X}=\mathbf{E}[X\mid Y] \tag{9.3.1}$$

as an estimator of X given Y, and the estimation error as

$$\widetilde{X}=X-\hat{X} \tag{9.3.2}$$

is a RV. ♡

Property 9.3 (Properties of the Estimator)

- *Unbiased:* $\mathbf{E}[\widetilde{X}]=\mathbf{E}[\mathbf{E}[\widetilde{X}\mid Y]]=\mathbf{E}[\mathbf{E}[X\mid Y]-\mathbf{E}[\hat{X}\mid Y]]=\mathbf{E}[\hat{X}-\hat{X}]=0$
- *Uncorrelated:* $\operatorname{cov}(\hat{X},\widetilde{X})=\mathbf{E}[\hat{X}\widetilde{X}]-\mathbf{E}[\hat{X}]\mathbf{E}[\widetilde{X}]=\mathbf{E}[\mathbf{E}[\hat{X}\widetilde{X}\mid Y]]=\mathbf{E}[\hat{X}\mathbf{E}[\widetilde{X}\mid Y]]=\mathbf{E}[\hat{X}\cdot 0]=0$
- $\operatorname{var}(X)=\operatorname{var}(\hat{X})+\operatorname{var}(\widetilde{X})+2\operatorname{cov}(\hat{X},\widetilde{X})=\operatorname{var}(\hat{X})+\operatorname{var}(\widetilde{X})$

The second to last equality of the second property is because $\hat{X}$ is fully determined by Y, i.e. given Y, $\hat{X}$ is a number.

The first item of the third property of the estimator $\operatorname{var}(\hat{X})$ can be written as

$$\operatorname{var}(\hat{X})=\operatorname{var}(\mathbf{E}[X\mid Y]) \tag{9.3.3}$$

The second item of the third property of the estimator $\operatorname{var}(\widetilde{X})$ can be written as

$$\operatorname{var}(\widetilde{X})=\mathbf{E}\left[\widetilde{X}^2\right]-\mathbf{E}\left[\widetilde{X}\right]^2=\mathbf{E}\left[\mathbf{E}\left[\widetilde{X}^2\mid Y\right]\right]=\mathbf{E}\left[\mathbf{E}\left[\left(X-\hat{X}\right)^2\mid Y\right]\right]=\mathbf{E}[\operatorname{var}(X\mid Y)] \tag{9.3.4}$$

So we get the total variance decomposition

Theorem 9.3 (Law of Total Variance)

$$\operatorname{var}(X)=\operatorname{var}(\mathbf{E}[X\mid Y])+\mathbf{E}[\operatorname{var}(X\mid Y)] \tag{9.3.5}$$ ♡

Appendix A Problem Solutions

A.1 Derived Distributions

In the lecture right after I finished this section, the lecturer provided an "official" solution to this problem, see section 8.1.

In the homework, I constantly encounter some problems like this:

Example A.1 (PDF of a Function of Multiple Continuous RVs) Let X, Y be two continuous RVs with joint PDF $f_{X,Y}(x, y)$, and $Z = g(X, Y)$. Find the PDF of Z.

Here I introduce two approaches to solve this kind of problems: CDF and Jacobian transformation.

Solution ***CDF:***

1. *Find the CDF of Z*

$$F_Z(z) = \mathbf{P}(Z \leq z) = \mathbf{P}(g(X, Y) \leq z) = \iint_{g(x,y)\leq z} f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y \tag{A.1.1}$$

2. *Differentiate $F_Z(z)$ to get the PDF of Z*

$$f_Z(z) = \frac{\mathrm{d}F_Z(z)}{\mathrm{d}z} \tag{A.1.2}$$

Solution ***Jacobian Transformation:*** *This approach requires finding a valid variable substitution with invertibility. Take $Z = g(X, Y) = X + Y$ as an example. Normally, if we want to calculate the determinant of the Jacobian matrix, the Jacobian matrix must be square. However, here Z is a scalar instead of a vector. We take the following steps to avoid this problem:*

1. *Let $U = X + Y$ and $V = X$, then $X = V$ and $Y = U - V$.*
2. *Calculate the Jacobian matrix and its determinant*

$$J = \begin{pmatrix} \frac{\partial X}{\partial U} & \frac{\partial X}{\partial V} \\ \frac{\partial Y}{\partial U} & \frac{\partial Y}{\partial V} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \tag{A.1.3}$$

$$\det(J) = -1 \tag{A.1.4}$$

3. *Find the joint PDF of U and V*

$$f_{U,V}(u, v) = f_{X,Y}(v, u - v) \cdot |\det(J)| = f_{X,Y}(v, u - v) \tag{A.1.5}$$

4. *Integrate out V to get the marginal PDF of U, which is the PDF of Z*

$$f_U(u) = \int_{-\infty}^{+\infty} f_{X,Y}(v, u - v)\,\mathrm{d}v \tag{A.1.6}$$